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NANYANG TECHNOLOGICAL UNIVERSITY

SEMESTER 1 EXAMINATION 2025-2026

EE2005 – ELECTRICAL DEVICES AND MACHINES

November / December 2025 Time Allowed: 2 hours

INSTRUCTIONS

1. This session paper contains 4 questions and comprises 4 pages.
2. Answer all the questions.
3. All questions carry equal marks.
4. This is a closed book examination.
5. Unless specifically stated, all symbols have their usual meanings.

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1. (a) Consider the magnetic circuit shown in Figure 1. It has three sections, namely, the left section, the center section and the right section. The structure has a uniform cross-sectional area of $A = 6.25$ cm^2 throughout. The dimensions of the structure are: $l_1 = 20$ cm, $l_2 = 5.95$ cm, $l_3 = 22$ cm, $l_g = 0.1$ cm. The coil wound on the left section consists of $N = 300$ turns and has a resistance of $R = 20\ \Omega$. The relative permeability of the core is $\mu_r = 2000$ and the permeability of free space is $\mu_o = 4\pi \text{x} 10^{-7}$ H/m.

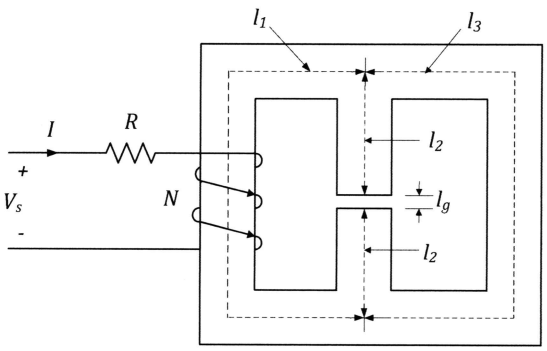


Figure 1

Note: Question No. 1 continues on page 2.

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(i) Draw the magnetic equivalent circuit of the magnetic structure shown in Figure 1 on page 1 and indicate all the reluctances of the various sections of the structure. If the DC supply voltage V_s is set to 90 V, determine the fluxes ϕ_1, ϕ_2 and ϕ_3 in the left, center and right sections, respectively. Ignore any leakage and fringing.

(ii) Determine the energy stored in each of the three sections and the percentage of the total energy stored in the air gap.

(iii) If the DC source is now replaced by a 50-Hz AC source, determine the amplitude of the source voltage to achieve half of the flux found in part (i).

(18 Marks)

(b) Figure 2 shows a relay with an air gap length of $l_g = 0.2$ cm. The coil wound on the magnetic core has a resistance of $R = 50\ \Omega$ and consists of $N = 200$ turns. The cross-sectional area is 4 cm^2. The relative permeability of the magnetic core and the movable armature is infinitely high. Assuming the gap faces are parallel, determine the force exerted on the armature when a DC supply voltage of $V_i = 20\ V$ is applied to the coil.

(7 Marks)

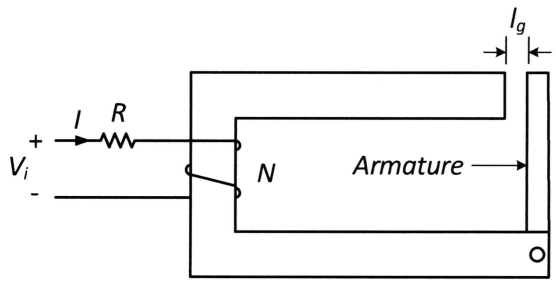


Figure 2

2. A single-phase transformer is rated 30 kVA, 2400/240-V, 50-Hz. The transformer has the following circuit parameter values:

$$R_1 = 0.225\ \Omega, \quad X_1 = 0.9\ \Omega, \quad R_2 = 0.0225\ \Omega, \quad X_2 = 0.09\ \Omega,$$

$$R_c = 13.5\ k\Omega, \quad X_M = 10.8\ k\Omega$$

(a) Draw the approximate equivalent circuit of the transformer referred to the primary side and hence, calculate the voltage regulation, efficiency and the input power factor if the transformer is operated at full load with a power factor of 0.85 lagging and at rated secondary voltage. Take the secondary voltage as the reference, i.e., $\vec{V}_2 = 240\angle 0^o$ V.

(15 Marks)

Note: Question No. 2 continues on page 3.

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(b) Suppose the transformer is now operated at 75 % full load with a power factor of 0.85 lagging and at rated secondary voltage. Assuming that the core losses are constant, determine the efficiency of the transformer and comment on your answer.

(3 Marks)

(c) The two-winding transformer is now connected as a step-down autotransformer. Select the connection that would lead to a large kVA rating. Draw the diagram of the step-down autotransformer, showing all the relevant voltages and currents and calculate the kVA rating of the autotransformer.

(7 Marks)

3. (a) The rotational losses and stator copper loss of a 460-V, 60-Hz, 6-pole three-phase induction motor are 3500 W and 3000 W, respectively. The torque developed by the motor is 700 Nm when it is running at a speed of 1152 rpm. Determine

(i) the input power of the motor, and
(ii) the output power of the motor.

(7 Marks)

(b) A three-phase, 3300-V, 60-Hz, 6-pole, wye-connected induction motor has the following per phase parameters referred to the stator:

$$R_1 = 0.5\ \Omega,\ R_2 = 0.5\ \Omega,\ X_M = 50\ \Omega,\ X_1 = 3\ \Omega,\ X_2 = 3\ \Omega$$

The rotational losses are 1000 W. Determine

(i) the speed at which maximum torque is developed,

(ii) the torque developed and the output power of the motor when it is running at a slip of 1 %, and

(iii) the resistance to be added in each phase of the rotor circuit, referred to the stator, such that the torque developed is 90 % of the value obtained in part (ii) and the motor is running at a slip of 5 %.

(18 Marks)

4. (a) A DC shunt motor has an armature winding resistance of 0.1 Ω and a field winding resistance of 130 Ω. The armature terminals are connected to a 130-V DC supply. At no load, the motor runs at a speed of 985 rpm and draws a current of 9 A from the DC supply. Determine

Note: Question No. 4 continues on page 4.

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(i) the rotational losses of the motor,

(ii) the armature current, input power and output power of the motor when the motor is driving a load at a speed of 900 rpm, and

(iii) the armature current and the value of the external resistance R_{ext} Ω inserted in series with the armature circuit if the speed of the motor is 750 rpm. Assume that the torque developed is the same as in part (ii).

(18 Marks)

(b) The armature winding resistance and field winding resistance of a separately-excited DC generator are 0.02 Ω and 70 Ω, respectively. The field circuit is connected to a 280-V DC supply. The generator is supplying a load current of 300 A to an external load at a terminal voltage of 300 V. If the efficiency of the generator at this load is 90 %, determine the rotational losses of the generator. Assume that the effect of armature reaction can be neglected.

(7 Marks)

END OF PAPER

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1. **Please do not turn over the question paper until you are told to do so. Disciplinary action may be taken against you if you do so.**

2. You are not allowed to leave the examination hall unless accompanied by an invigilator. You may raise your hand if you need to communicate with the invigilator.

3. Please write your Matriculation Number on the front of the answer book.

4. Please indicate clearly in the answer book (at the appropriate place) if you are continuing the answer to a question elsewhere in the book.